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IRIF – Paris

Formalization of Finite Elements in Coq

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Outline

1. Introduction
2. What is a Finite Element?
3. Property and Instanciation

Introduction

Mathematics

$\mathbb{R}$, $\int$, $\frac{\partial^2 u}{\partial t^2}$
theorems

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Applied Mathematics	numerical scheme, convergence algorithms + theorems

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Applied Mathematics	numerical scheme, convergence algorithms + theorems
Computer Science	floating-point numbers, implementation programs + ?

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Applied M

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Formal Proof

Motivations

PDE (Partial Differential Equations) $\Rightarrow$ weather forecast

$\Rightarrow$ nuclear simulation

$\Rightarrow$ optimal control

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$\Rightarrow$ real program that implements this scheme or this method

$\Rightarrow$ **towards verifying these programs!**

Motivations

http://www.ima.umn.edu/~arnold/disasters/sleipner.html

The sinking of the Sleipner A offshore platform

Excerpted from a report of SINTEF, Civil and Environmental Engineering:

The Sleipner A platform produces oil and gas in the North Sea and is supported on the seabed at a water depth of 82 m. It is a Condeep type platform with a concrete gravity base structure consisting of 24 cells and with a total base area of 16 000 m^2. Four cells are elongated to shafts supporting the platform deck. The first concrete base structure for Sleipner A sprang a leak and sank under a controlled ballasting operation during preparation for deck mating in Gandsfjorden outside Stavanger, Norway on 23 August 1991.

Immediately after the accident, the owner of the platform, Statoil, a Norwegian oil company appointed an investigation group, and SINTEF was contracted to be the technical advisor for this group.

The investigation into the accident is described in 16 reports...

The conclusion of the investigation was that the loss was caused by a failure in a cell wall, resulting in a serious crack and a leakage that the pumps were not able to cope with. The wall failed as a result of a combination of a serious error in the finite element analysis and insufficient anchorage of the reinforcement in a critical zone.

A better idea of what was involved can be obtained from this photo and sketch of the platform. The top deck weighs 57,000 tons, and provides accommodation for about 200 people and support for drilling equipment weighing about 40,000 tons. When the first model sank in August 1991, the crash





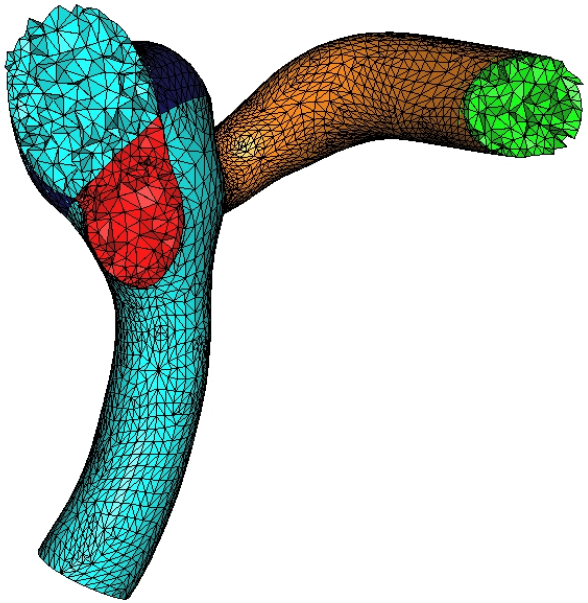
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Mesh



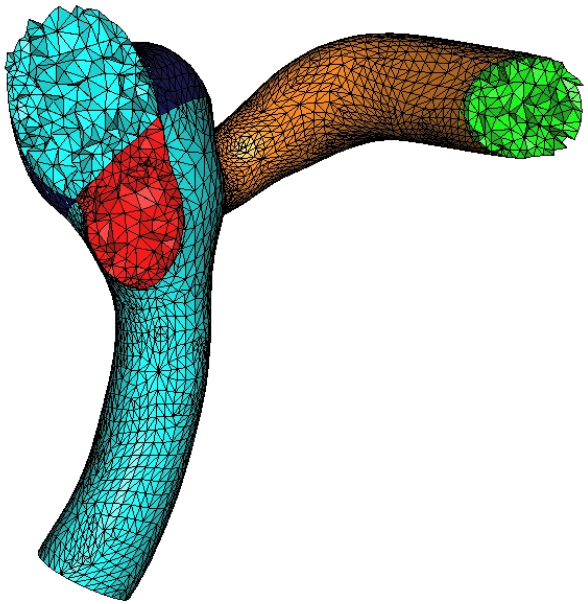
@ V. Martin

On a complex geometry, one has a mesh, made of triangles / tetrahedrons.

The Finite Element Method aims to approximate the PDE on each element, while constructing a continuous solution.

A Finite Element is a geometric element + ??? to solve the PDE on it.

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A **Finite Element** is how to approximate a function (solution of the PDE) by a simpler function (eg. polynomial) on a small simple geometric cell (eg. tetrahedron), by retaining particular values (eg. point values, mean values, fluxes through interfaces) to reduce to a finite-dimension problem.

Definition of a Finite Element

A Finite Element is a triple $(\mathcal{K}, \mathcal{P}, \Sigma)$:

1. $\mathcal{K}$ is a geometric cell, such as a simplex (segment in $\mathbb{R}$, triangle in $\mathbb{R}^2$, tetrahedron in $\mathbb{R}^3$)
2. $\mathcal{P}$ is a vector space of functions on $\mathcal{K}$, of dimension n_{dof} (often polynomials with bounded degree)
3. Σ is composed of n_{dof} linear forms on $\mathcal{P}$.

A Finite Element must have the unisolvence property:
For $\Sigma := \{\sigma_i\}_{i \in \{1:n_{dof}\}}$, let $\Phi_\Sigma : \mathcal{P} \to \mathbb{R}^{n_{dof}}$ be $\Phi_\Sigma(p) := (\sigma_i(p))_{i \in \{1:n_{dof}\}}$.
Unisolvence means that Φ_Σ is a bijection.

Corresponding Coq Definition

```
Record FE : Type := mk_FE {
  d    : nat ;                    (* space dimension eg 1, 2, 3 *)
  n_dof : nat ;                   (* nb of degrees of freedom *)
  d_pos :    0 < d ;              (* proof of positivity *)
  ndof_pos : 0 < n_dof ;          (* proof of positivity *)
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  shape : shape_type ;                          (* Simplex or Quad *)
  nvtx : nat := nvtx_of_shape d shape ;         (* number of vertices *)
  vtx : (ℝ^d)^nvtx ;                            (* vertices of geometrical element *)
  𝒦 : ℝ^d → Prop := convex_envelop vtx ;        (* geometrical element *)
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  P : (ℝ^d → ℝ) → Prop ;                        (* subset of functions *)
  P_approx_has_dim : has_dim  P  n_dof ;        (* its dimension is n_dof *)
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  Σ : (((ℝ^d → ℝ) → ℝ)^n_dof ;                  (* linear forms *)
  Sigma_lin_map : forall i, is_linear_mapping Σ_i ;
  unisolvence : bijS  P  fullset Φ_Σ ;
}.
```

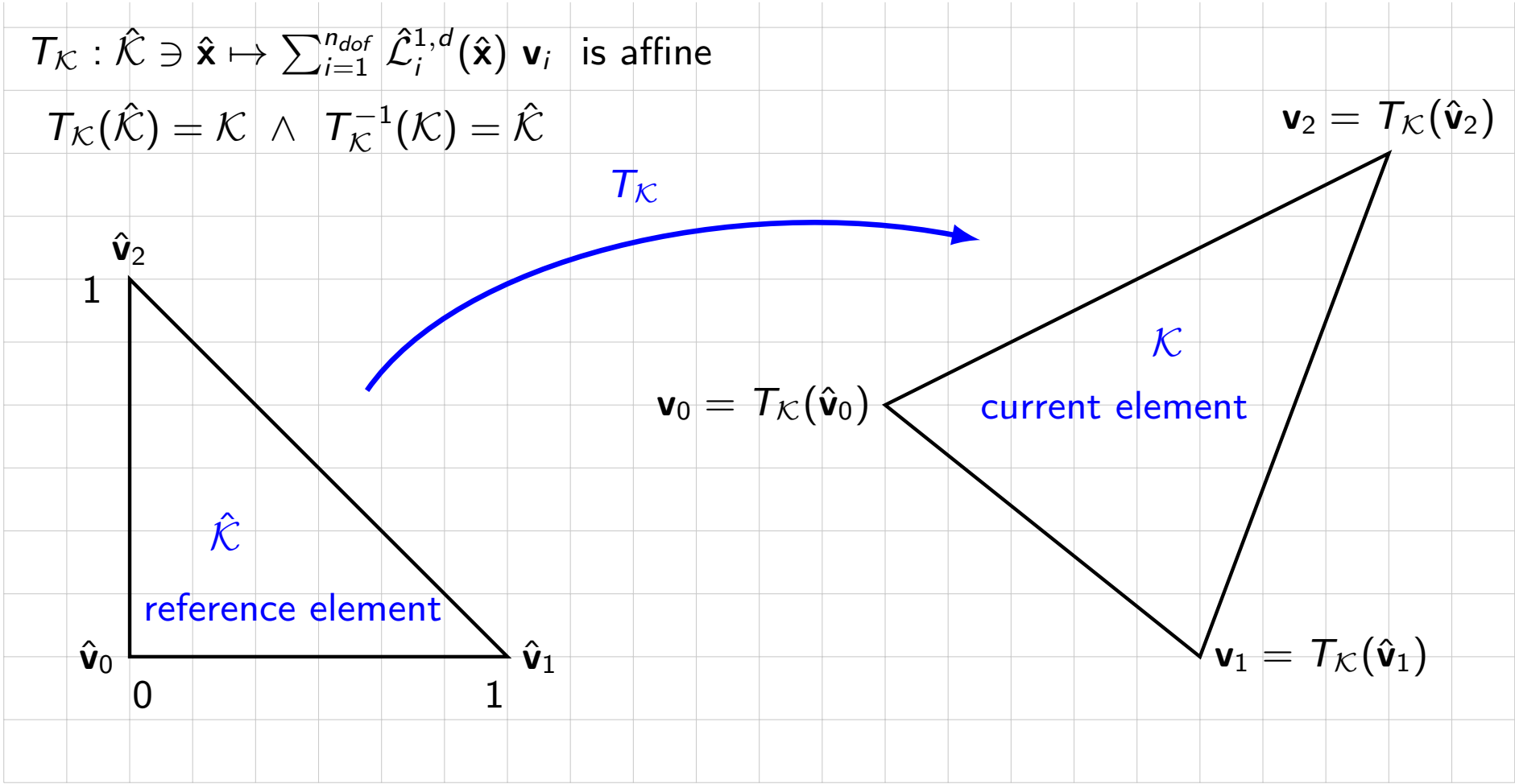
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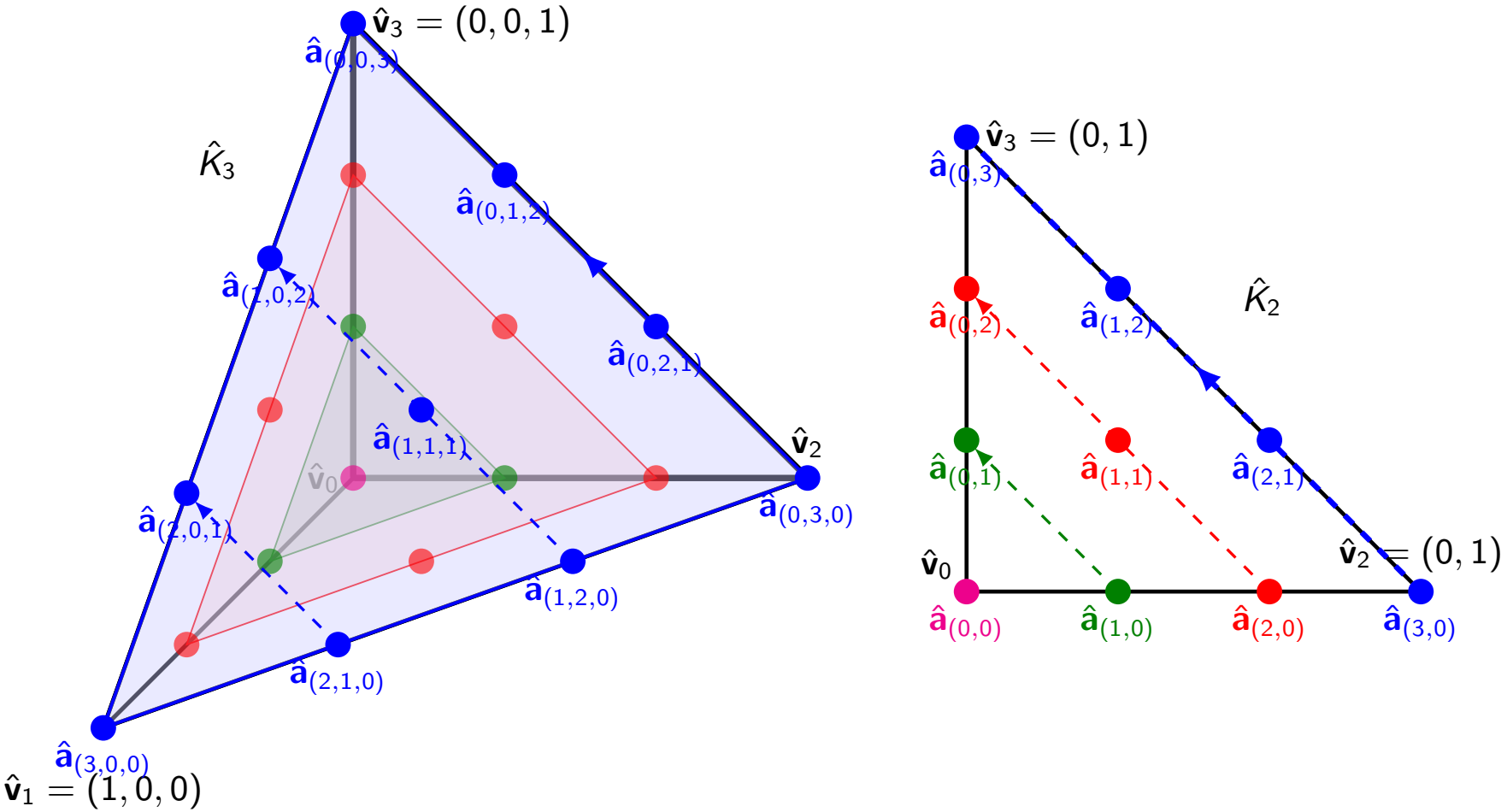
3 Property and Instanciation

Geometric transformation



From a reference finite element and the current vertices $\mathbf{v}_i$, we build a current finite element on K, relying on the geometric transformation T_K.

Definition of Lagrange Finite Elements 1/2



Reference tetrahedron and triangle with Lagrange nodes (evenly distributed).

Definition of Lagrange Finite Elements 2/2

Given d and k,

1. $\mathcal{K}$ is a simplex of dimension d (eg. triangle or tetrahedron),
2. $\mathcal{P}$ is the vector space of polynomials from $\mathbb{R}^d$ to $\mathbb{R}$ of degree $\leq k$,
3. the Σ_i are the values at the Lagrange nodes.

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```
Variable dL  kL : nat.

Hypothesis dL_pos : 0 < dL.

Variable vtx_cur : (ℝ^dL)^(dL+1).
Hypothesis Hvtx : affine_independent vtx_cur.

Definition FE_LagPdk_cur : FE :=
  mk_FE dL (nnode_LagPdk dL kL) dL_pos (nnode_LagPdk_pos dL kL)
  Simplex vtx_cur (Pdk dL kL) (Pdk_has_dim_ndof_LagPdk dL kL)
  Sigma_LagPdk_cur Sigma_LagPdk_cur_lm
  unisolvence_LagPdk_cur.
```

Conclusion

- interdisciplinary work: mathematics / computer science,
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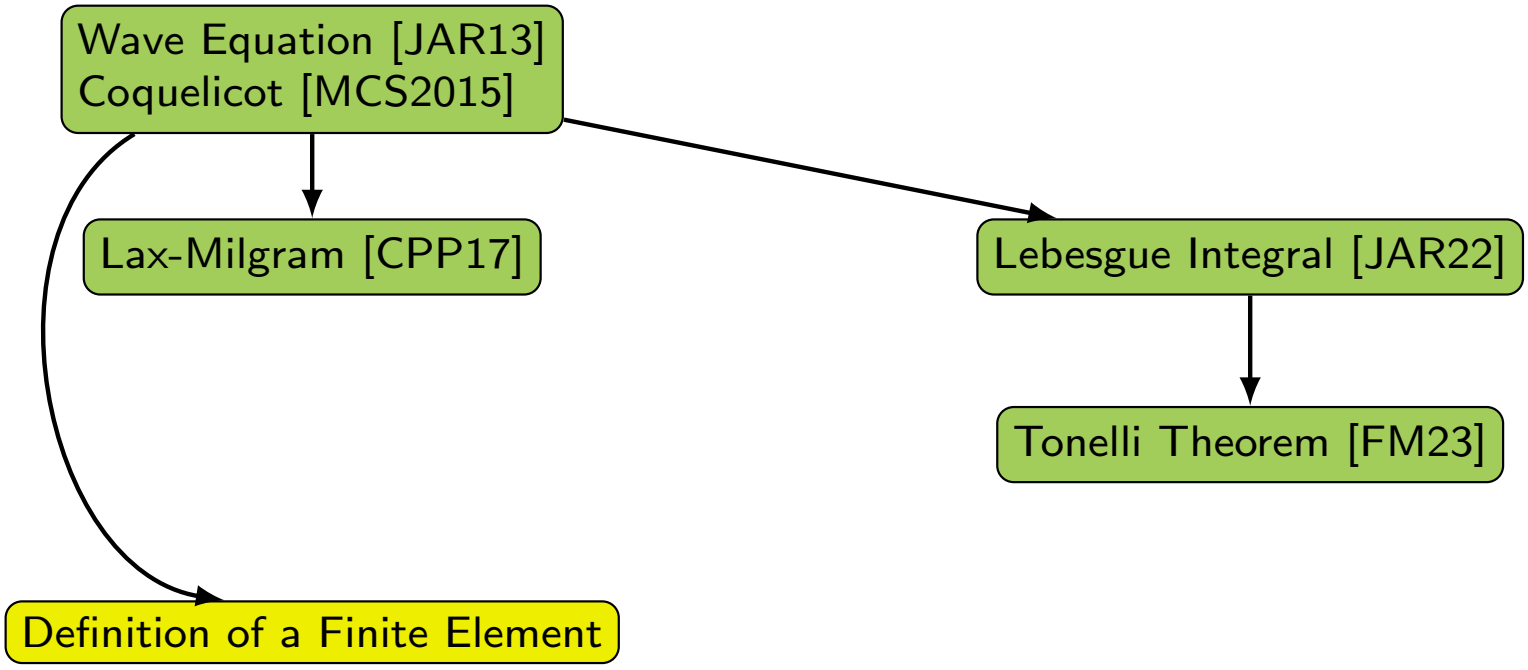
Difficulties:

- handling sub-spaces (with an algebraic structure, such as sub-groups or sub-vector spaces),
- handling finite families ("vectors") of the same size.

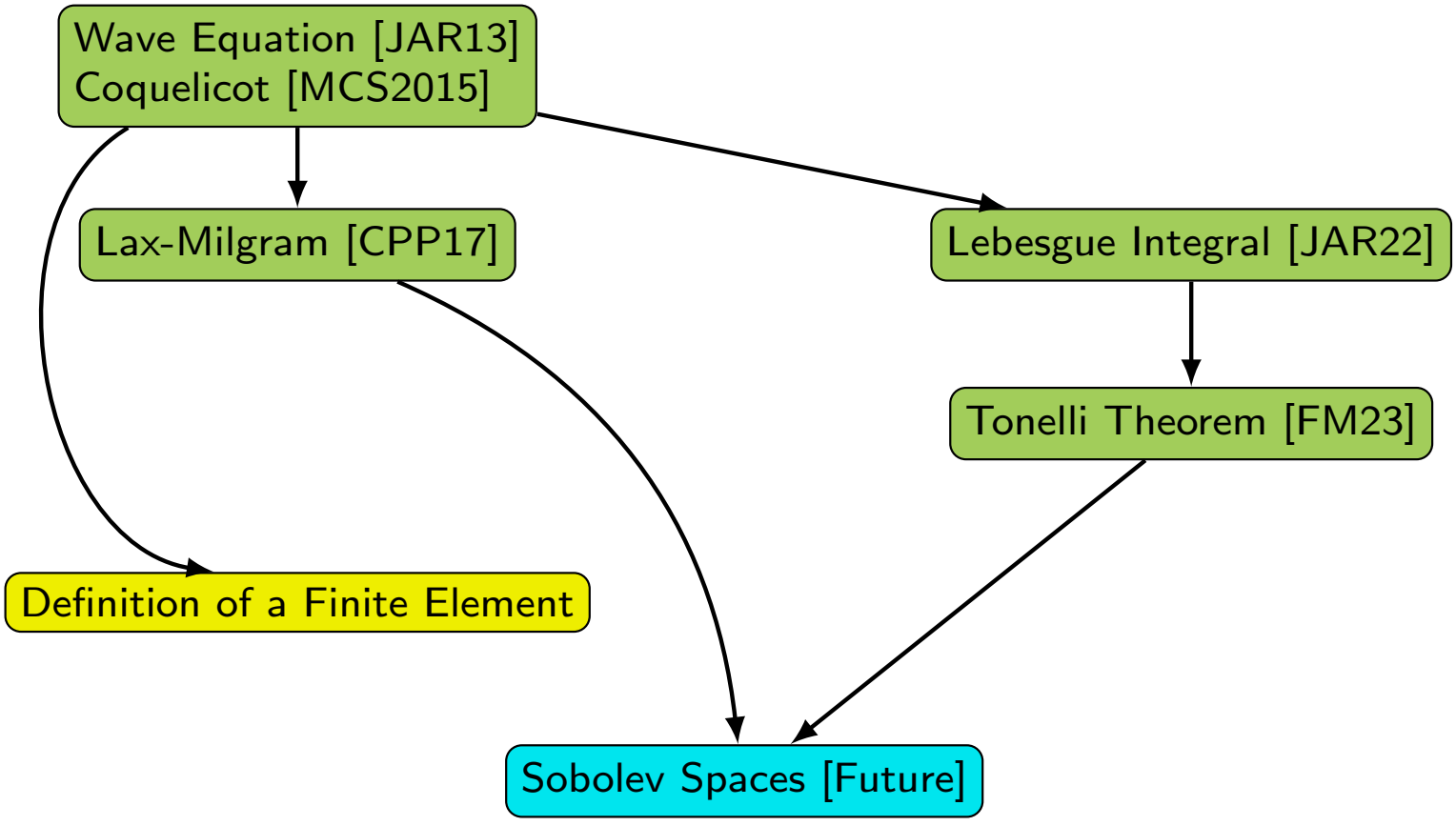
Perspectives

Definition of a Finite Element

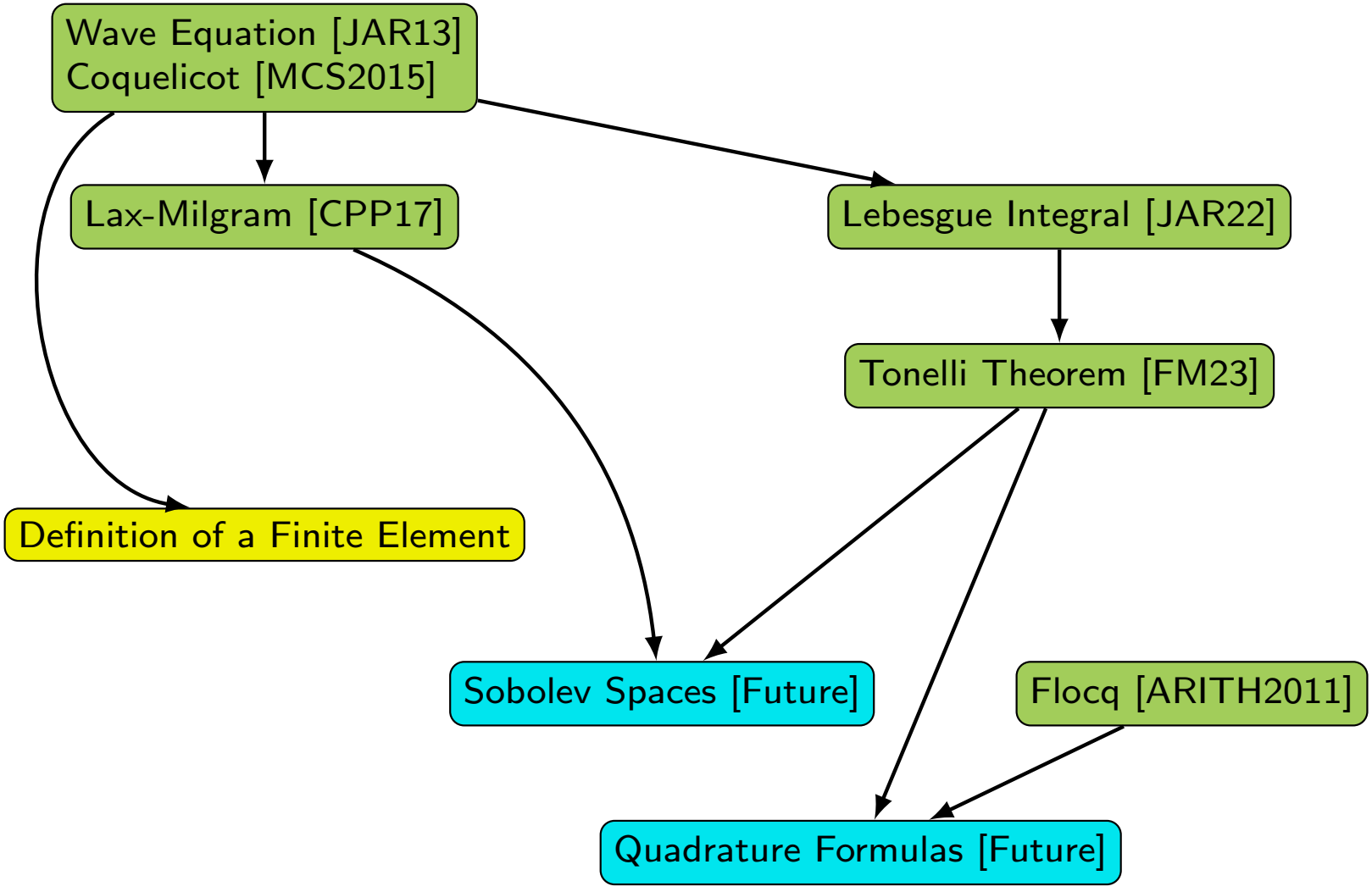
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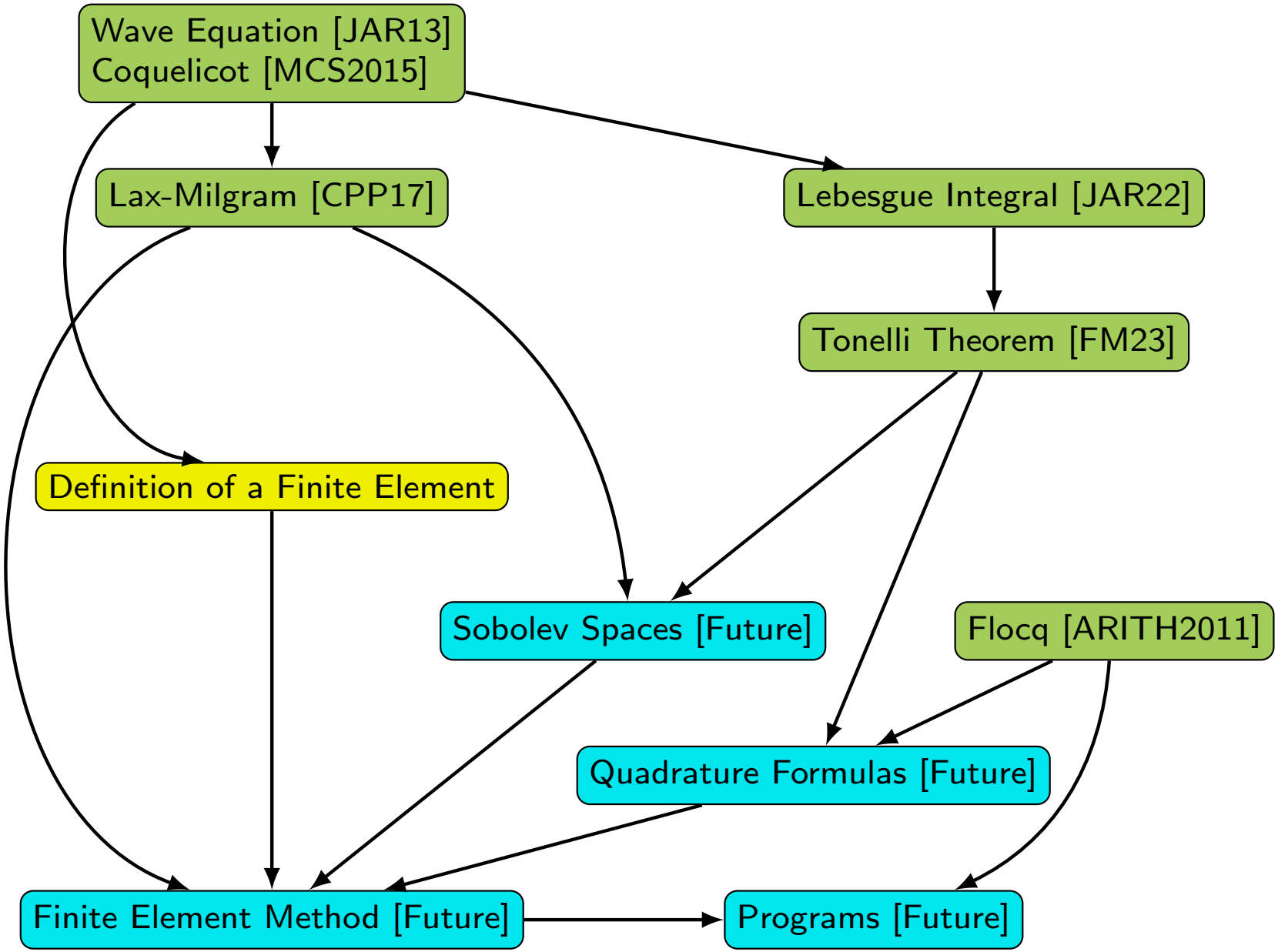
Perspectives



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Perspectives



$\mathcal{A}_d^k$

Given d and k, I want the vector of all the vectors of $\mathbb{N}^d$ with a sum smaller or equal to k.

- is this combinatorics?
- or geometry? (Lagrange nodes)
- or analysis? (monomials on $\mathbb{R}^d$ of degree $\leq k$).

We define this vector of vector called $\mathcal{A}_d^k$ (of size C_{d+k}^d) by concatenation.

```
Lemma Adk_sum : forall d k i, (sum (Adk d k i) ≤ k).

Lemma Adk_surj : forall d k (b:ℕ^d), (sum b ≤ k) → exists i, b = Adk d k i.

Lemma Adk_inj : forall d k, injective (Adk d k).

Lemma Adk_grsymlex_lt : forall d k, sortedF grsymlex_lt (Adk d k).
    (* a special order near the grevlex order on monomials *)
```